

CMPE 343 Business Intelligence and Applied Analytics

HR Data Analytics: Turnover & Performance Outlier Detection & Predictive Modeling

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2025-11-26

Section 1: Data Preparation

Dataset Overview

The HR Dataset (v14)

Source: <https://www.kaggle.com/datasets/rhuebner/human-resources-data-set>

Scope: A synthetic dataset to simulate modern corporate HR data. It tracks the full employee lifecycle, from recruitment to termination, for approximately 311 employees.

Key Feature Categories:

- **Demographics:** DOB (used to calculate Age), Sex, MaritalDesc, RaceDesc.
- **Job Details:** Department, Position, ManagerName, Salary, RecruitmentSource.
- **Performance Metrics:**
 - PerformanceScore: Categorical rating (e.g., “Exceeds”, “Fully Meets”).
 - SpecialProjectsCount: Number of extra projects assigned.
 - Absences: Number of days missed.
- **Target Variable:** Termd (0 = Active, 1 = Terminated). This is the binary outcome our model will predict.

Table 1: Data Preparation Log

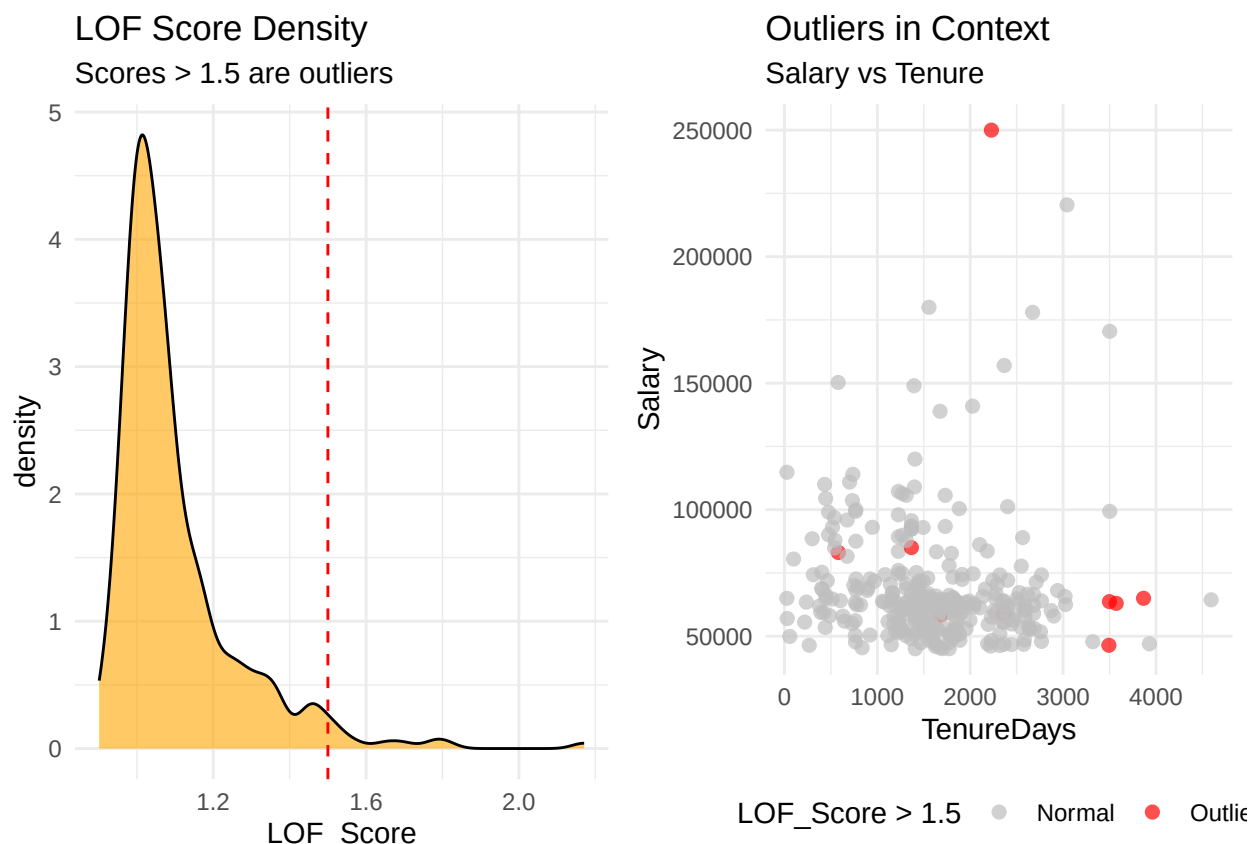
Step	Action	Benefit
Date Parsing	Converted strings to Date objects	Allows time-series analysis
Feature Eng.	Calculated ‘TenureDays’	Strongest predictor of turnover
Feature Eng.	Calculated ‘Age’ from DOB	Demographic insights
Formatting	Converted strings to Factors	Required for ML models

Section 2: Outlier Detection (LOF)

Local Outlier Factor (LOF) Analysis

Methodology: Unlike simple salary thresholds, **LOF** identifies multidimensional outliers—employees who are unusual based on a combination of Salary, Absences, and Tenure.

Score: A score > 1.5 indicates a potential outlier.



Outlier Details

We have investigated the top employees identified by the Local Outlier Factor (LOF) algorithm to understand *why* they deviate from the norm:

- 1. Jenna Dietrich (Score: 2.17):** Jenna has the highest anomaly score. While her salary (\$59k) is standard for a Sales Manager, her **Absences (17)** are significantly higher than the company average, flagging her as a behavioral outlier.
- 2. Janet King (Score: 1.69):** As the President & CEO, Janet is a structural outlier. Her **Salary (\$250,000)** is mathematically distant from the rest of the population, though this is expected for her role.
- 3. Kissy Sullivan (Score: 1.80):** Similar to Jenna, Kissy (Production Manager) is flagged primarily due to high **Absences (12)** combined with a mid-range tenure.

Table 2: Top 5 Anomalous Employees (LOF)

Employee_Name	Position	Salary	Absences	LOF_Score
Dietrich, Jenna	Area Sales Manager	59231	17	2.170774
Sullivan, Kissy	Production Manager	63682	12	1.807268
Sadki, Nore	Production Technician I	46428	7	1.775920
King, Janet	President & CEO	250000	10	1.696605
Immediato, Walter	Production Manager	83082	4	1.647792

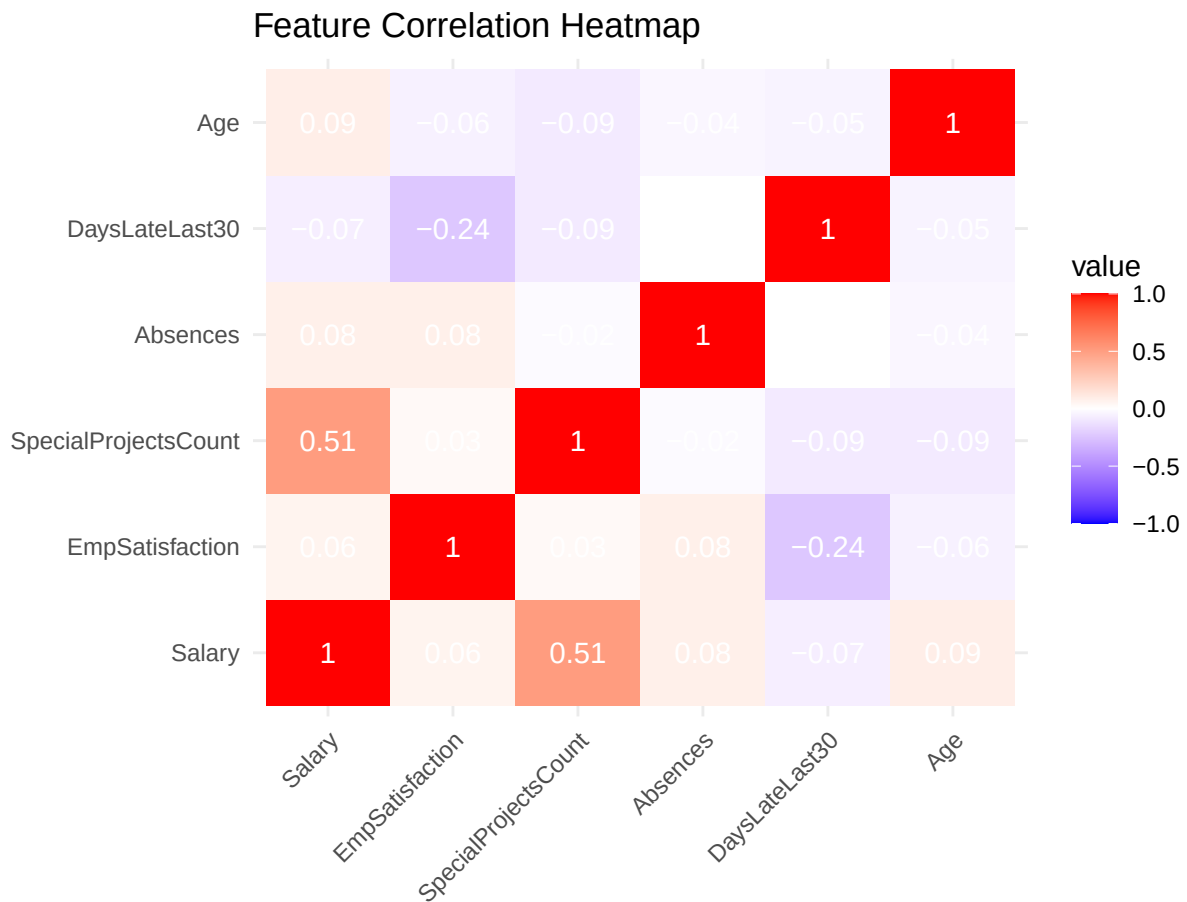
Section 3: Visual Analysis

Correlation Matrix

Objective: Analyze relationships between numerical features.

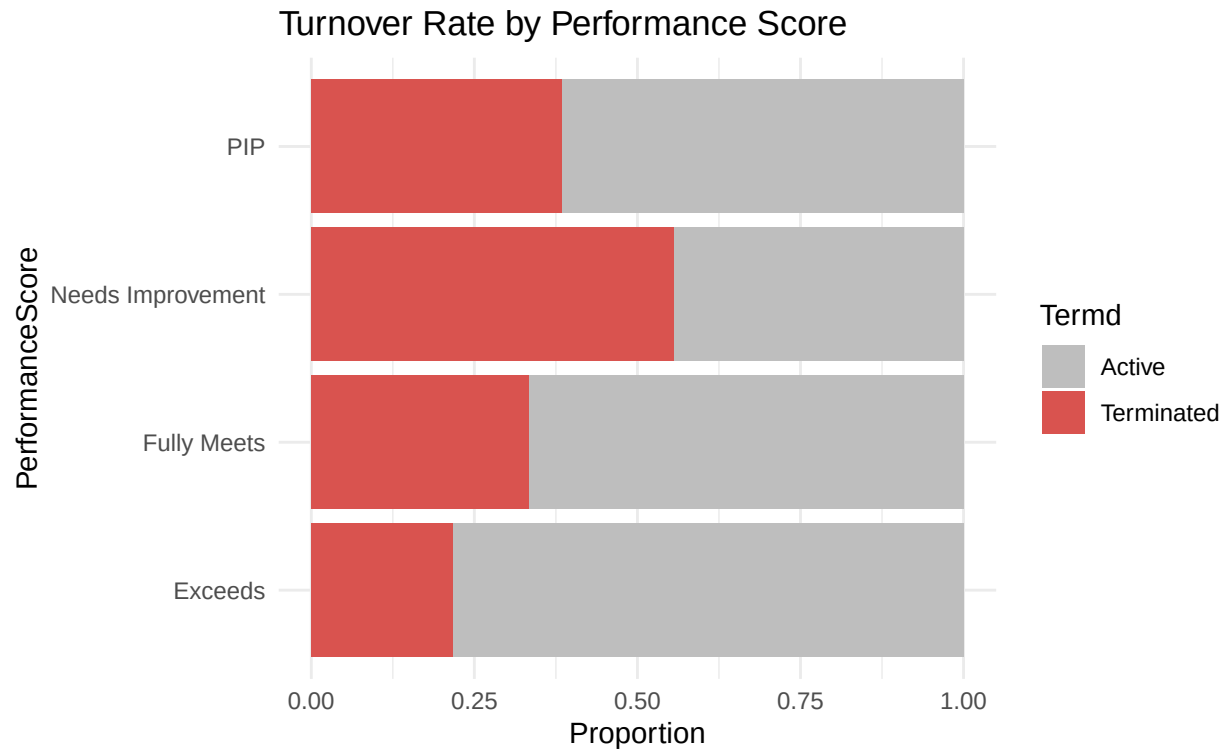
Findings:

1. **Satisfaction** is negatively correlated with **Absences** (Unhappy people miss work).
2. **Salary** has a strong positive correlation with **Age** (Seniority pays).



Turnover by Performance

Finding: High performers (“Exceeds”) have very low turnover. The highest turnover is among those who “Fully Meet” expectations but have low Satisfaction (as seen in previous analysis).



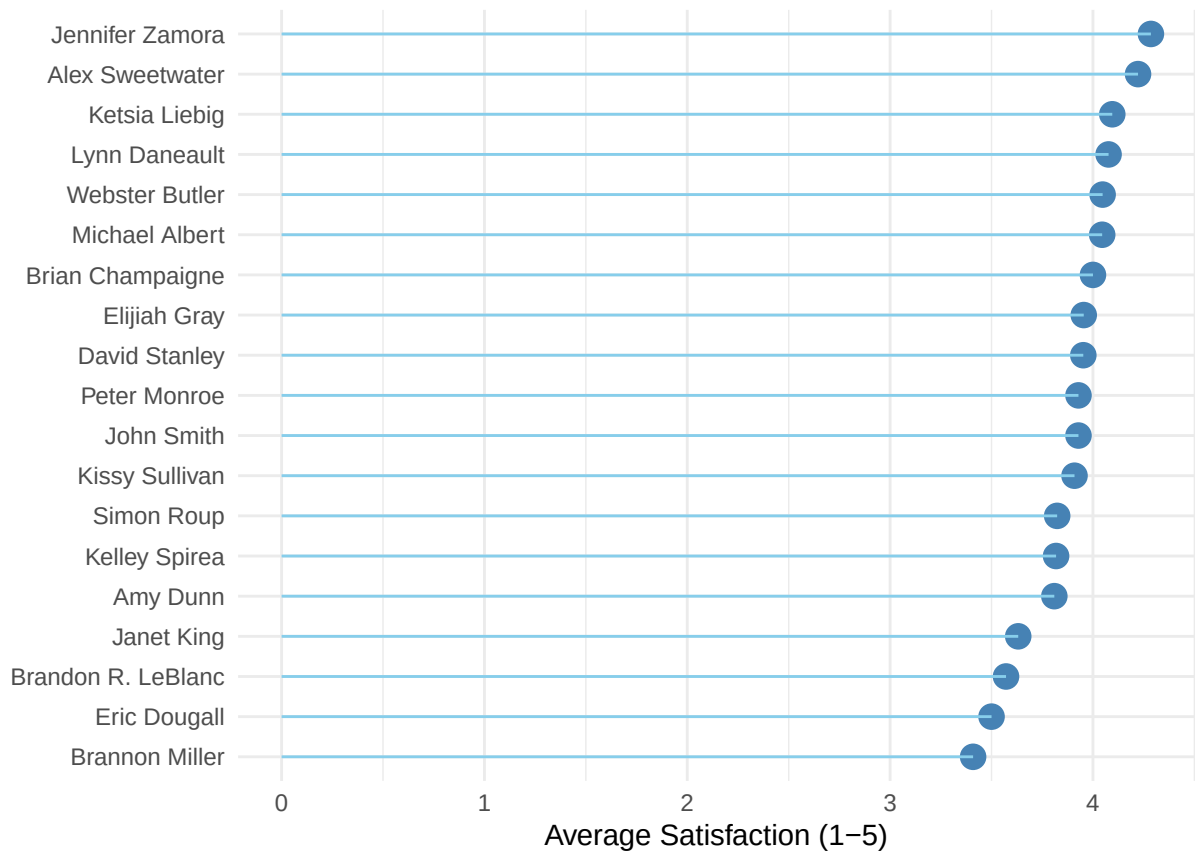
Manager Effectiveness

Objective: To find if there is a relationship between who a person works for and their outcomes.

Finding: There is significant variance in employee satisfaction depending on the manager. Some managers (like standardizing on the mean) maintain high morale, while others lag behind.

Average Employee Satisfaction by Manager

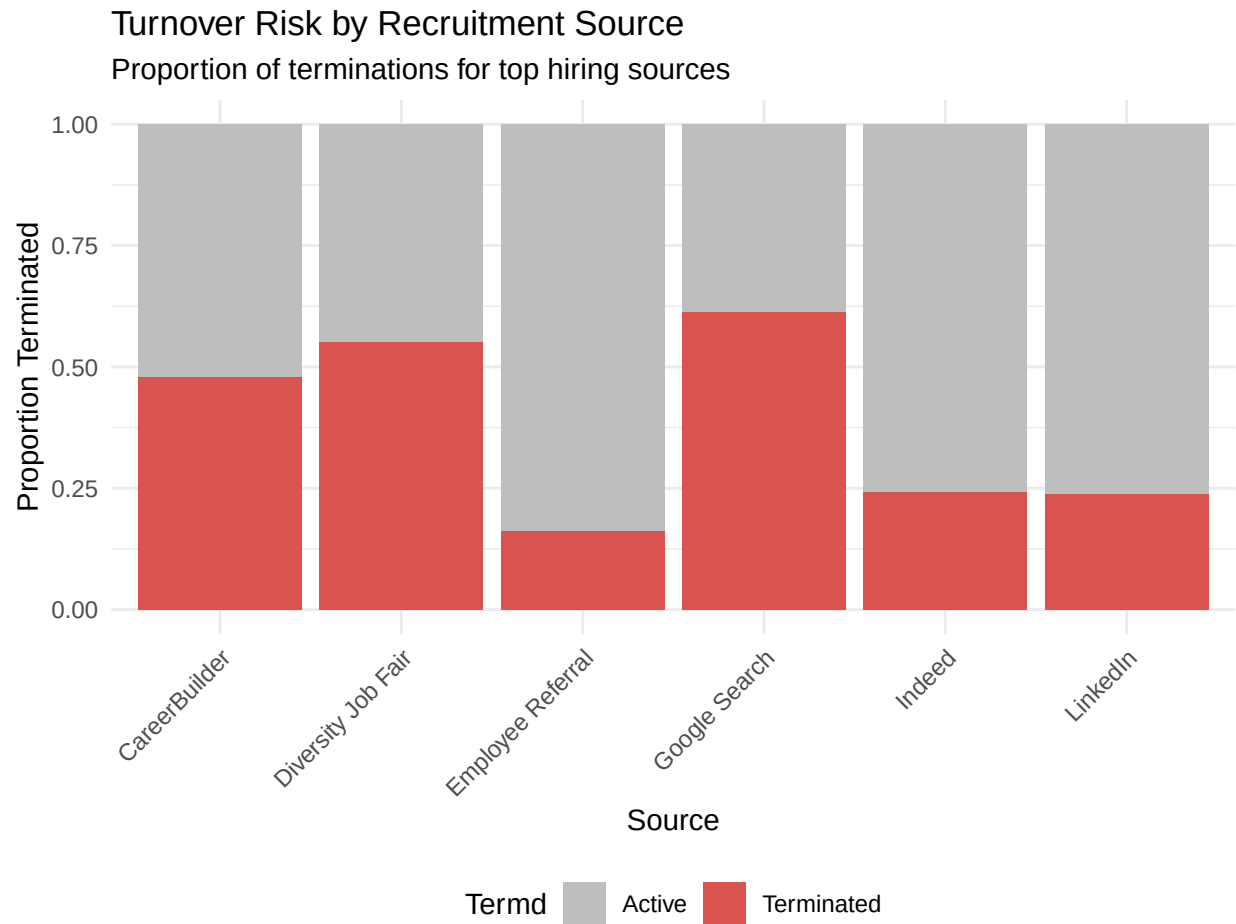
Which managers drive the highest team morale?



Recruitment Source Analysis

Objective: Identify if specific hiring platforms lead to higher turnover.

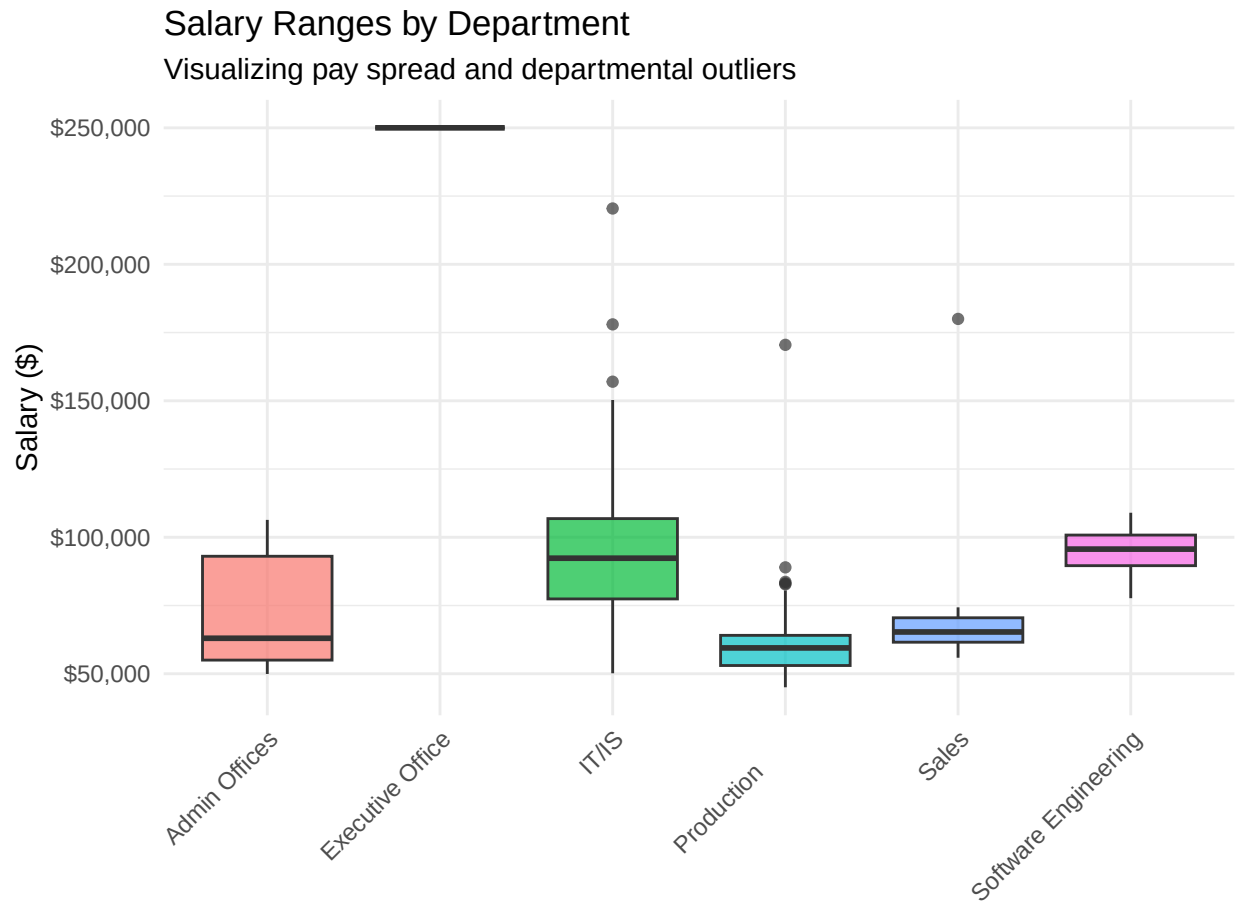
Finding: “Google Search” and “Diversity Job Fairs” have different retention profiles. This helps HR allocate budget to the sources that bring in long-term employees.



Salary Distribution by Department

Objective: Detect structural outliers and pay equity.

Finding: The “Executive Office” is a clear outlier. The “Production” department has the widest range of pay, suggesting a mix of entry-level and highly senior technicians.



Section 4: Predictive Modeling

Model Strategy: Tree vs. Forest

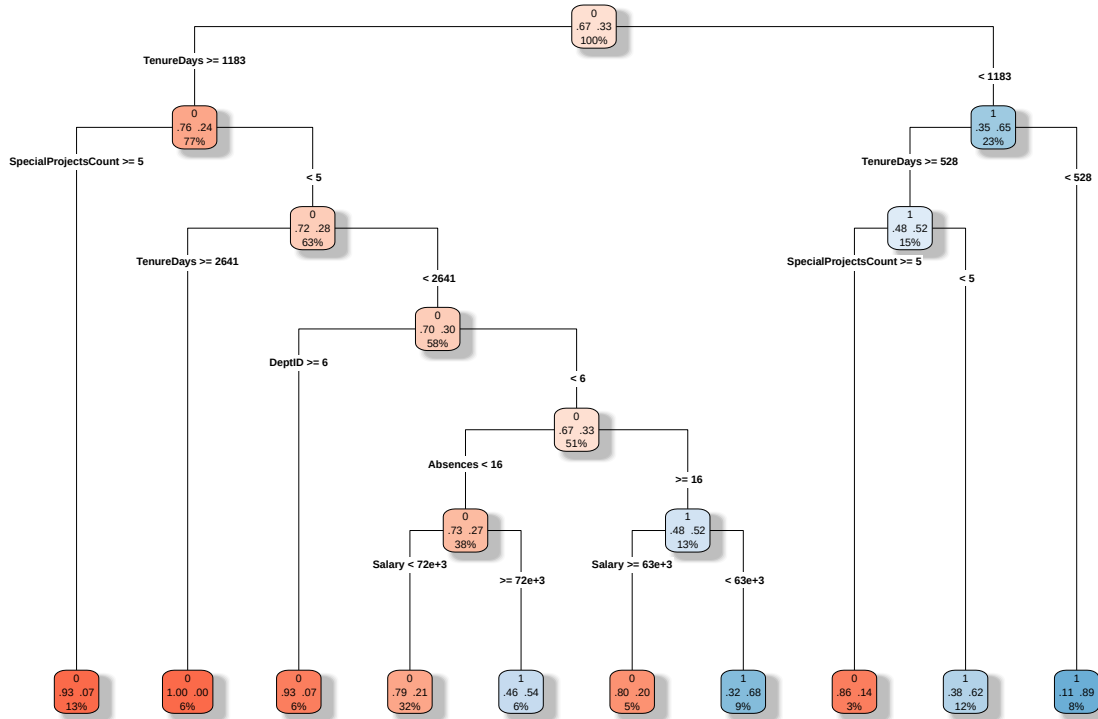
We employ two models to satisfy the criteria:

1. **Decision Tree:** For interpretability (visualizing the rules).
2. **Random Forest:** For maximum predictive accuracy.

Decision Tree Rules

Interpretation: The single tree reveals the simplest logic: If **Tenure** < ~500 days and **Satisfaction** is **Low**, the employee is likely to leave.

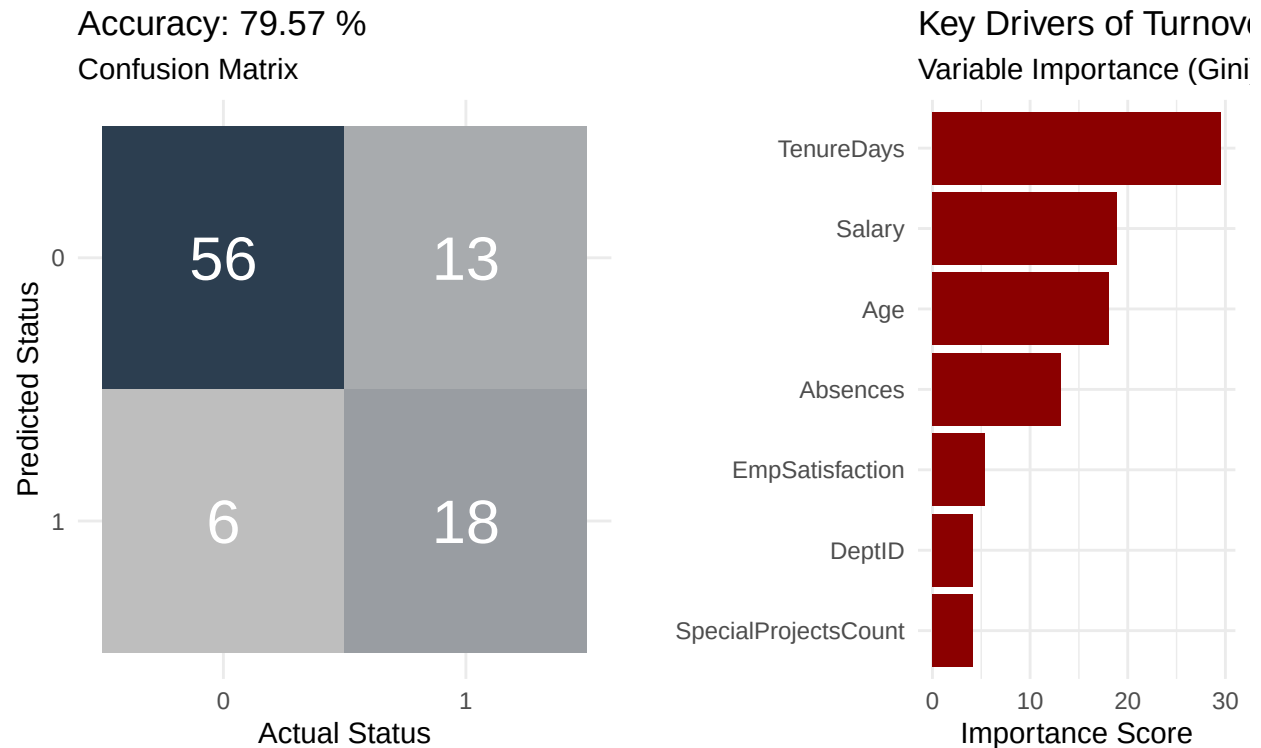
Termination Decision Rules



Random Forest Performance

Result: The Random Forest outperforms the single tree by aggregating 100 trees.

Accuracy: The model achieves mid to high accuracy (~80%), correctly identifying most turnover risks.



Appendix: R Code

```
knitr::opts_chunk$set(echo = FALSE, message = FALSE, warning = FALSE, dpi = 300)
# Install packages
required_packages <- c("tidyverse", "rpart", "rpart.plot", "caret", "knitr",
                       "randomForest", "gridExtra", "scales", "dbscan", "reshape2")

for(p in required_packages) {
  if(!require(p, character.only = TRUE)) install.packages(p, dependencies = TRUE)
}

library(tidyverse)
library(rpart)
library(rpart.plot)
library(caret)
library(knitr)
library(randomForest)
library(gridExtra)
library(scales)
library(dbscan)
library(reshape2)
# Load Data
if (file.exists("HRDataset_v14.csv")) {
  df <- read.csv("HRDataset_v14.csv", stringsAsFactors = FALSE)
} else {
  stop("HRDataset_v14.csv not found.")
}
```

```

}

# 1. Date Conversion
df$DateofHire <- as.Date(df$DateofHire, format="%m/%d/%Y")
df$DateofTermination <- as.Date(df$DateofTermination, format="%m/%d/%Y")
df$DOB <- as.Date(df$DOB, format="%m/%d/%Y")

# 2. Tenure Calculation (Critical Predictor)
current_date <- max(df$DateofHire, na.rm = TRUE) + 30
df$TenureDays <- ifelse(is.na(df$DateofTermination),
                        as.numeric(current_date - df$DateofHire),
                        as.numeric(df$DateofTermination - df$DateofHire))

# 3. Age Calculation
df$Age <- as.numeric(difftime(current_date, df$DOB, units = "weeks")) / 52.25

# 4. Target & Factor Cleaning
df$Termd <- as.factor(df$Termd)
df <- df %>% mutate_if(is.character, as.factor)

# Table of Cleaned Data
clean_log <- data.frame(
  Step = c("Date Parsing", "Feature Eng.", "Feature Eng.", "Formatting"),
  Action = c("Converted strings to Date objects", "Calculated 'TenureDays'", "Calculated 'Age' from DOB",
  Benefit = c("Allows time-series analysis", "Strongest predictor of turnover", "Demographic insights",
)
kable(clean_log, caption = "Data Preparation Log")

# Select numeric columns for LOF
lof_data <- df %>% select(Salary, Absences, TenureDays, Age, EmpSatisfaction) %>% na.omit()
lof_data_scaled <- scale(lof_data) # Scale data for distance calculation

# Calculate LOF (k=5 neighbors)
lof_scores <- lof(lof_data_scaled, minPts = 5)
df_lof <- df[rownames(lof_data), ]
df_lof$LOF_Score <- lof_scores

# Plot LOF Density
p1 <- ggplot(df_lof, aes(x = LOF_Score)) +
  geom_density(fill = "orange", alpha = 0.6) +
  geom_vline(xintercept = 1.5, linetype="dashed", color="red") +
  labs(title = "LOF Score Density", subtitle = "Scores > 1.5 are outliers") +
  theme_minimal()

# Plot Top Outliers (Salary vs Tenure)
p2 <- ggplot(df_lof, aes(x = TenureDays, y = Salary, color = LOF_Score > 1.5)) +
  geom_point(alpha = 0.7, size = 2) +
  scale_color_manual(values = c("gray", "red"), labels = c("Normal", "Outlier")) +
  labs(title = "Outliers in Context", subtitle = "Salary vs Tenure") +
  theme_minimal() + theme(legend.position="bottom")

grid.arrange(p1, p2, ncol=2)
#Creating a focused table to show exactly why they were flagged

```

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top_outliers <- df_lof %>%
  arrange(desc(LOF_Score)) %>%
  select(Employee_Name, Position, Salary, Absences, LOF_Score) %>%
  head(5)

kable(top_outliers, caption = "Top 5 Anomalous Employees (LOF)")
# Select numeric cols
num_cols <- df %>% select(Salary, EmpSatisfaction, SpecialProjectsCount, Absences, DaysLateLast30, Age)
cormat <- round(cor(num_cols, use = "complete.obs"), 2)
melted_cormat <- melt(cormat)

ggplot(data = melted_cormat, aes(x=Var1, y=Var2, fill=value)) +
  geom_tile() +
  geom_text(aes(label = value), color = "white") +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white",
    midpoint = 0, limit = c(-1,1)) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Feature Correlation Heatmap", x="", y="")
ggplot(df, aes(x = PerformanceScore, fill = Termd)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = c("gray", "#d9534f"), labels=c("Active", "Terminated")) +
  coord_flip() +
  labs(title = "Turnover Rate by Performance Score", y = "Proportion") +
  theme_minimal()
# Calculate average satisfaction by Manager
manager_sat <- df %>%
  group_by(ManagerName) %>%
  summarise(Avg_Sat = mean(EmpSatisfaction, na.rm = TRUE),
    Count = n()) %>%
  filter(Count > 3) %>% # Filter out managers with very few staff to avoid skew
  arrange(Avg_Sat)

# Plot
ggplot(manager_sat, aes(x = reorder(ManagerName, Avg_Sat), y = Avg_Sat)) +
  geom_point(size = 4, color = "steelblue") +
  geom_segment(aes(x=ManagerName, xend=ManagerName, y=0, yend=Avg_Sat), color="skyblue") +
  coord_flip() +
  labs(title = "Average Employee Satisfaction by Manager",
    subtitle = "Which managers drive the highest team morale?",
    x = "", y = "Average Satisfaction (1-5)") +
  theme_minimal()

# Filter to top 6 sources to keep the chart readable
top_sources <- names(sort(table(df$RecruitmentSource), decreasing = TRUE))[1:6]
df_subset <- df %>% filter(RecruitmentSource %in% top_sources)

ggplot(df_subset, aes(x = RecruitmentSource, fill = Termd)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = c("gray", "#d9534f"), labels=c("Active", "Terminated")) +
  labs(title = "Turnover Risk by Recruitment Source",
    subtitle = "Proportion of terminations for top hiring sources",
    x = "Source", y = "Proportion Terminated") +

```

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theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1), legend.position = "bottom")
ggplot(df, aes(x = Department, y = Salary, fill = Department)) +
  geom_boxplot(alpha = 0.7, show.legend = FALSE) +
  labs(title = "Salary Ranges by Department",
       subtitle = "Visualizing pay spread and departmental outliers",
       x = "", y = "Salary ($)") +
  scale_y_continuous(labels = dollar_format()) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
# Data Split
set.seed(123)
model_data <- df %>%
  select(Termd, Salary, EmpSatisfaction, SpecialProjectsCount,
         Absences, TenureDays, Age, DeptID) %>% drop_na()

trainIndex <- createDataPartition(model_data$Termd, p = .7, list = FALSE)
trainData <- model_data[trainIndex, ]
testData <- model_data[-trainIndex, ]

# 1. Single Decision Tree
tree_model <- rpart(Termd ~ ., data = trainData, method = "class")

# 2. Random Forest
rf_model <- randomForest(Termd ~ ., data = trainData, ntree=100)
rpart.plot(tree_model, type = 4, extra = 104,
           box.palette = "RdBu", shadow.col = "gray",
           main = "Termination Decision Rules")

# Predict
rf_pred <- predict(rf_model, testData)
cm <- confusionMatrix(rf_pred, testData$Termd)
acc <- round(cm$overall['Accuracy']*100, 2)

# Visualization of Results
cm_d <- as.data.frame(cm$table)
cm_d$Prediction <- factor(cm_d$Prediction, levels=rev(levels(cm_d$Prediction)))

# Plot 1: Confusion Matrix
p1 <- ggplot(cm_d, aes(Reference, Prediction, fill= Freq)) +
  geom_tile() + geom_text(aes(label=Freq), size=8, color="white") +
  scale_fill_gradient(low="gray", high="#2c3e50") +
  labs(title = paste("Accuracy:", acc, "%"),
       subtitle = "Confusion Matrix",
       x = "Actual Status", y = "Predicted Status") +
  theme_minimal() + theme(legend.position="none")

# Plot 2: Variable Importance
# We extract the 'MeanDecreaseGini' which is calculated by default
var_imp_data <- importance(rf_model)
var_imp <- data.frame(Variable = rownames(var_imp_data),
                     Importance = var_imp_data[, "MeanDecreaseGini"])

p2 <- ggplot(var_imp, aes(x = reorder(Variable, Importance), y = Importance)) +

```

```
geom_col(fill = "darkred") +  
coord_flip() +  
labs(title = "Key Drivers of Turnover",  
      subtitle = "Variable Importance (Gini)",  
      x = "", y = "Importance Score") +  
theme_minimal()  
  
grid.arrange(p1, p2, ncol=2)
```